4.9 Lighting Devices

Lighting devices may be used in work zones when specified in this Manual or by the Road Authority. They may supplement channelizing devices, signs, and barriers, and/or be used to illuminate equipment or work activity areas.

4.9.1 Flashing Vehicle Lights

All work, buffer, and shadow vehicles stationed in or near travel lanes should be equipped with 4-way (emergency) flashers and 360-degree flashing yellow lights. They should be activated whenever a vehicle is positioned such that it could influence traffic.

4.9.2 Yellow Warning Lights

Yellow warning lights are portable yellow lights commonly used during night time hours to supplement other traffic control devices.

There are four types of yellow warning lights—Types A, B, C, and D. Types A and B should not normally be mixed when used in a series.



Figure 4.23: Type A & C Yellow Warning Light

- **Type A Low-Intensity** flashing warning lights may be used to supplement signs and channelizing devices such as barriers, barricades (particularly Type 1), and flexible drums, and for marking specific hazards and outlining long-duration work activity areas.
- **Type B High-Intensity** flashing warning lights are used during daylight and night-time hours. They operate 24 hours per day and may be mounted on advance warning signs or on independent supports. They may also be used on barricades, barriers, and channelizing devices if they can be dimmed during night-time use to reduce driver sensitivity to brightness.

  Type B lights should be **visible for at least 300 metres** in clear **daylight** conditions. Flags may be substituted for Type B lights in day-time applications.
- **Type C Steady-Burn and Type D 360-Degree Steady Burn** warning lights may be used to supplement delineation of the edge of the roadway where multiple devices are used. They may also be used to delineate a work activity area or to run along the edge of a closed lane or shoulder. When used to delineate a curve, they should be used only on devices on the outside of the curve.



Figure 4.24: Type D Yellow Warning Light

All light types should be **visible for at least 900 metres** under clear night-time conditions.

Warning lights may be mounted on signs or channelizing devices and shall be secured in such a manner that they are unlikely to penetrate the windshield if hit by an errant vehicle.

Warning lights can operate in either flashing or steady-burn mode:

- Flashing warning lights are not to be used for delineation because a series of flashers operating randomly does not identify the desired vehicle path.
- For tapers, a series of sequential synchronized flashing warning lights or steady burn flashers may be placed on channelizing devices in order to increase driver detection and recognition.
- If a series of sequential flashing warning lights is used, the successive flashing should occur from the upstream end to the downstream end of the taper in order to identify the desired vehicle path.
- Each flashing warning light in a sequence should be flashed at a rate no less than 55 times per minute and no more than 75 times per minute.

4.9.3 Roadway Lighting

Consider the temporary installation of luminaires at key locations in very long-duration work zones. Areas that may benefit from the installation of roadway lighting include:

- project staging areas
- road hazards (e.g., structure encroachments)
- crossovers
- diversions (bypasses)
- areas with sudden alignment changes
- curves
- intersections
- transitions from multilane divided roadways to two-lane, two-way roadways

4.9.4 Overhead Lighting

Overhead lighting includes floodlights, balloon lights, and existing street lighting. When work is performed at night, overhead lighting should be used to illuminate the work activity area, equipment crossings, and other potentially hazardous areas. Overhead lighting is not a substitution for the need for retroreflectivity on signs and devices.

Except in emergency situations and for mobile operations, each Traffic Control Person location shall be illuminated at night by overhead lighting. More information can be found in *Section 5.4.1 TCP Apparel and Equipment*.

The adequacy of overhead lighting placement and glare elimination should be determined by driving through and observing the lit area at night—from each direction on all approaching roadways—both after the initial overhead lighting is set up and periodically thereafter.

Floodlights may have to be shielded or repositioned to prevent glare. Floodlighting of some areas may decrease relative visibility in other areas, and it may be necessary to define intended vehicular paths with other devices, such as strings of steady-burn yellow warning lights.

4.9.5 Equipment Lighting

Equipment lighting is critical for the visibility of equipment on the road and in work activity areas. LED lighting is becoming the preferred lighting option for equipment. Other lighting options are still acceptable.

All powered mobile equipment shall be equipped with lighting to ensure that it is visible to drivers, pedestrians, and workers. Equipment lighting requirements include 360-degree flashing lights and 4-way yellow flashing lights or equivalent for 360-degree visibility.

Other visibility devices that may be used or required on work zone equipment include:

- retroreflective striping
- equipment-mounted lamps for localized area lighting
- multiple lights mounted around equipment to light the work activity area

To reduce glare, balloon lighting may be used for lighting equipment. It can also be effective for night-time paving operations.



Figure 4.25: Balloon Lighting

4.10 Fencing and Screens

4.10.1 Work Zone Fencing

Work zone fencing can be used to:

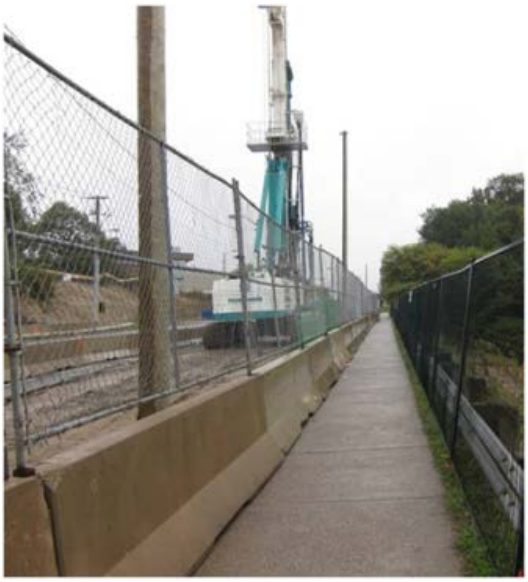
- identify the work area
- protect the public from the work area
- protect the work area from road users
- mark the edge of work zones
- identify the travel path for drivers, cyclists, and pedestrians



Removable Chain Link Panels



Snow Fence with Retroreflective Stripes



Chain Link Panels on Concrete Roadside Barrier

Figure 4.26: Work Zone Fencing

4.10.2 Work Zone Screens and Barrier Screens

In long-duration work zones on multilane highways where median and roadside work activities may impact traffic operations and cause delays, screens are useful for blocking the road user's view of activities, which can be distracting.

Screens may further improve safety and traffic flow by reducing headlight glare from oncoming vehicle traffic.

Screens may be mounted on the top of temporary traffic barriers that separate two-way vehicle traffic. They shall not be mounted where they might adversely affect vehicle operations or driver sight distances.

4.11 Other Traffic Control Devices

4.11.1 Flags on Traffic Control Devices

Flags are used to enhance the daylight visibility of certain traffic control devices in speed zones of 70 km/h or higher. They are fluorescent red or orange squares at least 40 cm x 40 cm (16" x 16") in size. They are not required for night work because their effectiveness is limited by poor lighting conditions. Flags used on signs during the day may be replaced with Type A flashing lights at night.

Flags should be used in pairs and positioned so as not to interfere with the visibility of the sign messages. They shall not be used by Traffic Control Persons to direct traffic.

Flags are used on signs that warn of day-time workers on or adjacent to a roadway with regular posted speed limits ≥ 70 km/h:

- Traffic Control Person Ahead C-001-1
- Survey Crew Ahead C-003
- Crew Working Ahead C-004
- Accident Scene C-058

Flags should generally be used only on the signs listed above, which relate directly to the presence of workers. They may also be used on other signs that require additional emphasis—for example, layouts requiring few signs but covering long distances (mowing, line markings, etc.).

Note: 1. Flags should not be used on all signs in a sign series because overuse reduces the emphasis and effectiveness of key signs.

2. Flags should not be used on speed limit signs.

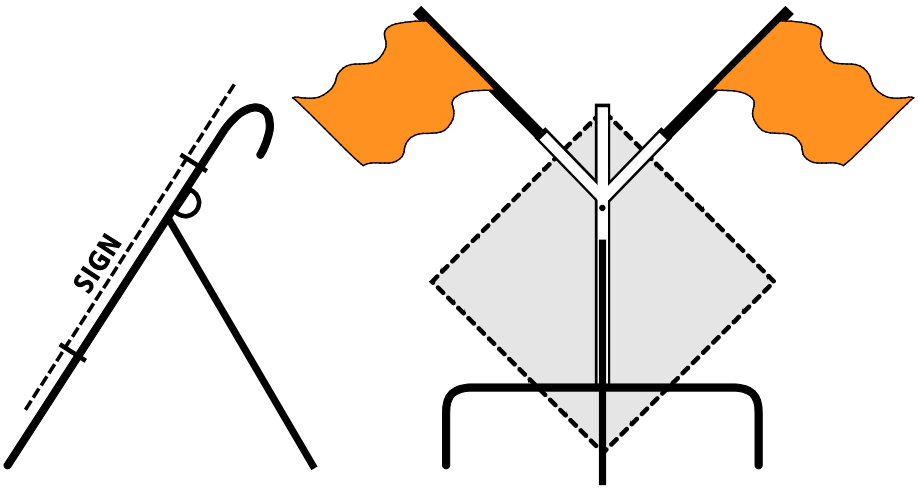


Figure 4.27: Pair of Flags on Sign

4.11.2 Sand Bags/Weights

Sand bags/weights may be used in work zones to support and/or stabilize the base area of signs and channelizing devices. Windy areas, higher-speed roadways, and narrow and sloped areas impact the stability of traffic control devices. The weights should be placed near the road surface where they cannot become projectiles.

Sand bags shall not be used as a channelizing device.

4.11.3 Speed Reader Boards

Speed reader boards (SRB) are electronic changeable speed display signs capable of detecting and displaying the speed of approaching vehicles in real-time via radar speed detection. SRB come either as trailer mounted units or pole mounted units. SRB may be used for:

- long-duration work zones (i.e., night-time work or more than one day-time shift)
- work zones that use Traffic Control Persons
- highway projects when stipulated by provisions in the project documents

1. Deployment Guidelines

1. SRBs should be positioned downstream of the regulatory speed sign to affirm a construction speed zone.
2. SRBs are placed approximately 100 to 200 metres in advance of the work activity area.
3. When used in advance of Traffic Control Person setups, the location of the speed reader board should follow the placement guidelines illustrated in *Figure 4.29 A: Long-Duration Lane Closure with TCPs and Speed Reader Boards – Two-Lane, Two-Way Roadway.*
4. On multilane highways, speed reader boards should not be placed close to merge areas and ramp areas.
5. Where work zones are divided into several work activity areas or are more than 1500 metres long, two or more speed reader boards may be used (per direction) to reaffirm and maintain speed reductions, separated by at least 300 metres.
6. Speed reader boards should be inspected for sight lines and for shadowing created by structures or construction signs to ensure sign visibility and effective operation.
7. A speed reader board should be delineated/protected using drums or tube markers/ delineators, with at least three markers/delineators on the upstream side of the speed reader board.

2. Operational Guidelines

1. SRB should include the words YOUR SPEED or similar text, together with the numeric electronic display. The text may be non-electronic.
2. SRB shall be in operation only when the construction speed limit is in effect and workers are present on the roadway.
3. The speed reader board shall be programmed in relation to the construction speed limit.
4. If no vehicles are approaching the speed reader board, the display should be blank.
5. The electronic display may be programmed to flash and/or display the message SLOW DOWN when the vehicle speed exceeds 10 km/h over the speed limit. It shall not flash for speeds less than 10 km/h over the speed limit. The flash rate shall be a maximum of 50 cycles per minute. Strobe-type light enhancements are not permitted.
6. When the vehicle speed exceeds 40 km/h over the speed limit, the numeric display should be programmed to go blank or display the message SLOW DOWN.
7. The radar in the speed reader board should detect an approaching vehicle no more than 10 seconds before the vehicle reaches the radar unit's position. Detection should not occur until the vehicle has entered the construction speed zone.



Figure 4.28: Speed Reader Board

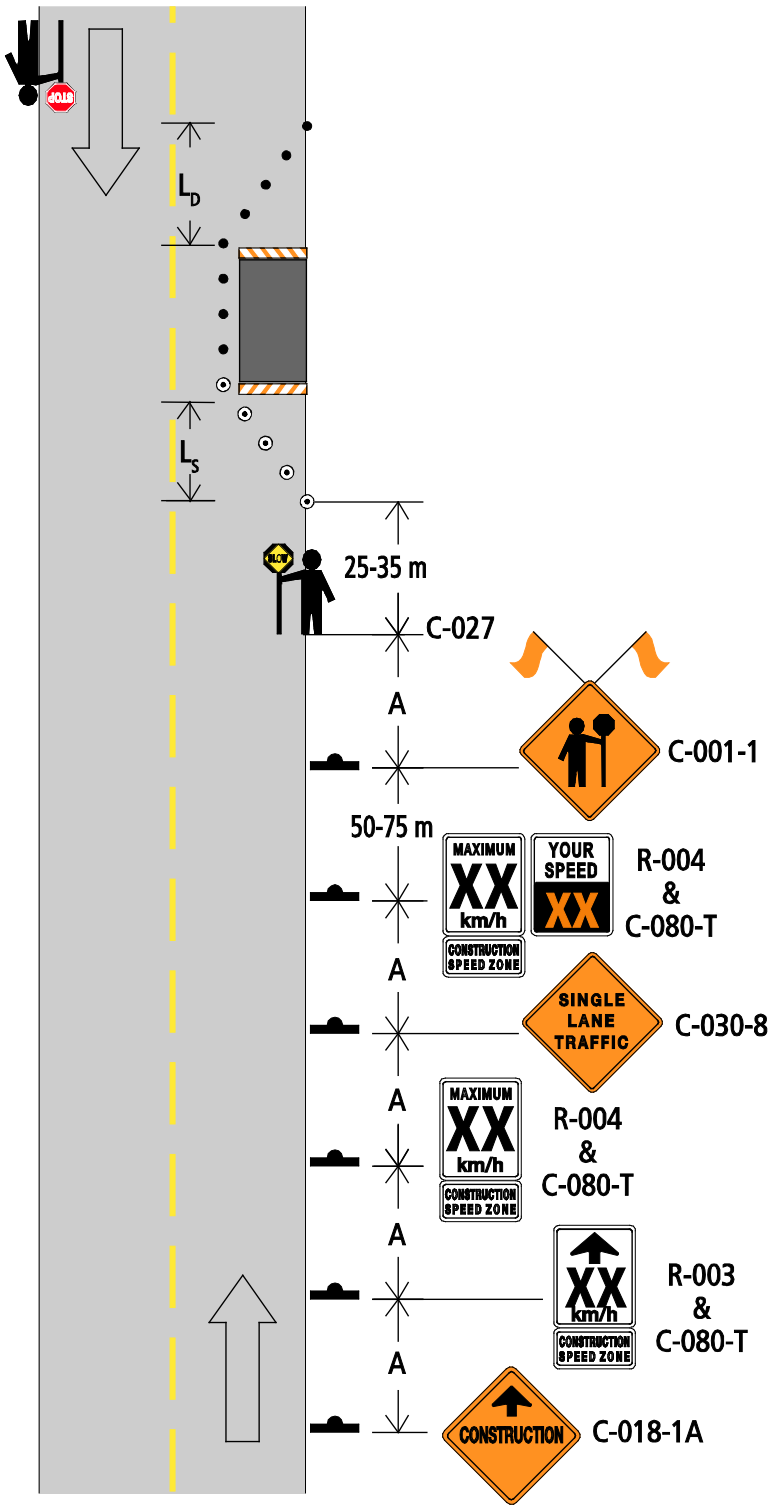


Figure 4.29 A: Long-Duration Lane Closure with TCPs and Speed Reader Boards – Two-Lane, Two-Way Roadway

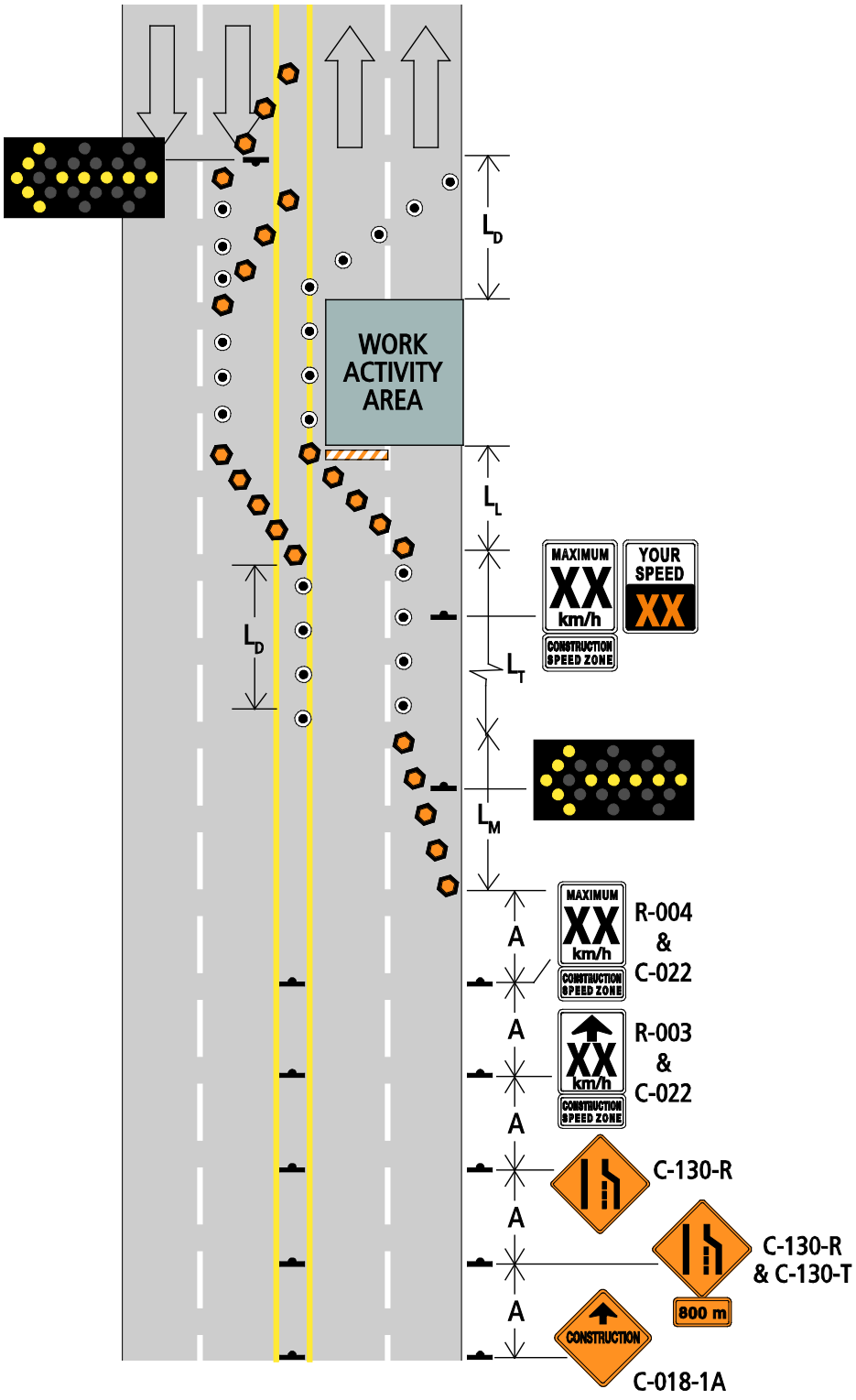


Figure 4.29 B: Freeway/Multilane Median Crossover with Speed Reader Board

4.11.4 Temporary Rumble Strips

Temporary rumble strips may be used to alert road users to a changing roadway environment that requires extraordinary caution. They may also be used as an audible vehicle detection system for workers adjacent to the roadway.

They are surface placed, raised strips, which are placed perpendicular to the direction of travel. When a vehicle passes over the strips, the noise and vibration draw the driver's attention to features such as signs, unexpected alignment changes, or potential stop conditions.

Installation Guidelines:

1. Spacing between temporary rumble strips should be 3.0 m, and their width should extend across the travel lane. A sign warning drivers of the rumble strips should be placed in advance of the installation (see *Figure 4.31: Layout of Temporary Rumble Strips*).
2. Temporary rumble strips may be white, yellow, black, or orange, and contrast the colour of the roadway.
3. Temporary rumble strips should be placed sufficiently in advance of the condition to allow road users to respond to the warning.
4. Temporary rumble strips should not be placed:
    - within intersections
    - through pedestrian crossings
    - on sharp horizontal or vertical curves
    - within marked bicycle lanes or on roadways used by cyclists unless a clear path at least 1.0 m wide is provided at each edge of the roadway or on each paved shoulder



Figure 4.30: Temporary Rumble Strips

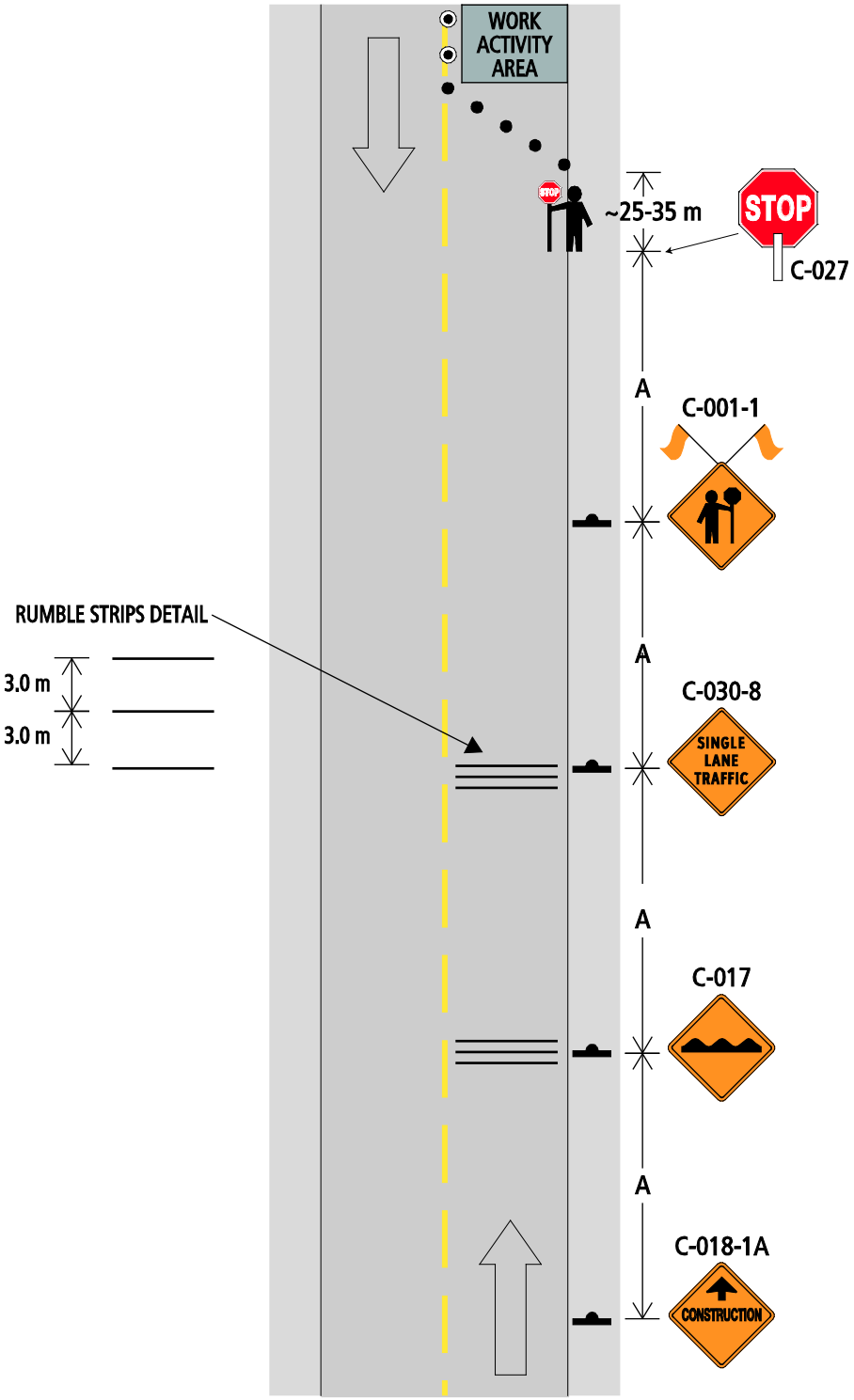


Figure 4.31: Layout of Temporary Rumble Strips

4.11.5 Shadow Vehicles

Shadow vehicles are used to provide mobile advance warning for operations where a work vehicle blocks or encroaches into a travel lane that has not been closed to traffic.

There may be more than one shadow vehicle for a continuously-moving work zone, with one shadow vehicle positioned as far as possible onto the shoulder (left or right) in advance of the work vehicle. Two shadow vehicles are typically required on multilane divided roadways with speeds ≥ 70 km/h. The shadow vehicle operator(s) and the work vehicle operator should be in communication with one another.

Operations where shadow vehicles may be used include, but are not limited to:

- pavement marking and striping
- hydro-seeding
- sweeping
- flushing
- pothole patching
- mowing

The factors involved in determining the requirement for shadow vehicles include:

- exposure of workers to traffic
- speed of traffic relative to speed of work vehicle
- traffic volumes and number of lanes
- highway classification
- shoulder width
- sight distance
- weather conditions

Shadow vehicle distances should be adjusted for horizontal and vertical curves so that the vehicle is clearly visible to traffic approaching the curves. The distance between shadow vehicles and working equipment may require periodic adjustment to prevent drivers from crossing or driving into the lane between the shadow vehicle and the work area or equipment.

Shadow vehicles shall be equipped with a flashing arrow board (FAB) and a 360-degree flashing yellow light and 4-way flashers. If a flashing arrow board is used on a two-lane, two-way roadway, it should show only a non-directional warning display. It shall never display an arrow that directs traffic into a lane that could be occupied by opposing traffic.

The shadow vehicle may be equipped with a rear-mounted crash attenuator. This may be required for specific types of work activities for certain highway classifications, or as specified by the Road Authority. Vehicle-mounted crash attenuators are often used on shadow vehicles in mobile, high-speed (≥ 70 km/h) operations.

4.11.6 Buffer Vehicles

Buffer vehicles are stationary vehicles used to protect workers from errant vehicles in an active work area. The buffer vehicle is parked upstream of the workers. The wheels should be pointed in a direction that will help to prevent the vehicle from entering the work activity area or travel lanes if it is struck.

Buffer vehicles shall be equipped with a 360-degree flashing yellow light and 4-way flashers, or a flashing arrow board (FAB). They may also be equipped with vehicle-mounted crash attenuators to reduce the effect of a collision.

4.11.7 Vehicle-Mounted Crash Attenuators

Vehicle-mounted crash attenuators are energy-absorbing devices attached to the rear of shadow vehicles, buffer vehicles, or trailers to help protect workers or equipment from errant vehicles. Their energy-absorption properties help to reduce crash severity.



Figure 4.32: Trailer-Style Crash Attenuator

Vehicle-mounted crash attenuators may be used in many applications that require a buffer vehicle or additional protection for workers and the work zone. They are often used on shadow vehicles in mobile, high-speed (≥ 70 km/h) operations. The Road Authority may define in the contract those situations that require their use.



Figure 4.33: Vehicle-Mounted Crash Attenuator

4.11.8 Temporary Crash Attenuators on Barriers

Crash attenuators—also known as crash cushions or impact attenuators—are systems that absorb energy when struck by an errant vehicle, either through deceleration or deflection. These devices reduce the effects of crashes from the exposed ends of barriers, bridge piers, fixed objects, and other obstacles. Detailed crash attenuator information is available in the American Association of State Highway and Transportation Officials' (AASHTO) Roadside Design Guide.

Requirements for the use of temporary crash attenuators are defined in the Ministry's special provisions for highways projects, or by the Road Authority.

1. Crash attenuators used on Provincial highways shall meet the current American American Association of State Highway and Transportation Officials (AASHTO) Manual for Assessing Safety Hardware (MASH) unless otherwise specified by the Road Authority .
2. Unless otherwise approved by the Road Authority, the selected test level shall match the original or intended regulatory speed at which the highway will operate when the work zone is removed.
3. Crash attenuators should be inspected periodically to verify that they have not been hit or damaged. Damaged crash attenuators shall be repaired or replaced to maintain their crash-worthiness.
4. For Ministry projects, the Regional Traffic Engineer will assess the need for crash attenuators based on several factors. These include, but are not limited to:
    - speed
    - highway classification
    - number of lanes
    - volume
    - geometrics
    - site constraints
    - obstacle to be protected

4.11.9 Pilot Cars for Work Zones

Section 4.11.9 deals with pilot cars that are used to guide traffic through construction zones and work areas.

For information on piloting extraordinary loads, see the following:

- *Appendix G: Pilot Car Load Movement Guidelines*
- Division 8 (Pilot Cars and Signs) in the *Commercial Transport Regulations* http://www.bclaws.ca/civix/document/id/complete/statreg/30_78
- Chapter 8 of the *Commercial Transport Procedure Manual* https://www2.gov.bc.ca/assets/gov/driving-and-transportation/cvse/commercial-transportation-manual/chapter-8.pdf

For the purposes of this Manual, a pilot car is a vehicle marked with warning signs and lights that is used to guide a queue of vehicles through a work zone or detour regulated by Traffic Control Persons or by temporary signals for which the pilot car operator has full control of the signal operation. The length and complexity of the work zone makes navigation difficult for drivers. (e.g., where there is a substantial change in alignment).

When deciding whether or not to use a pilot car operation, it is important to consider the type of work, traffic volume, road alignment, and access points within the work zone.

A pilot car should have four or more wheels, seating for two or more persons, and be capable of transporting pedestrians or cyclists through the work zone.

1. Communication with Stakeholders

It is important to communicate with the affected stakeholders when initiating a pilot car operation in an area with accesses and driveways via:

- written notification of the dates and times when work will take place
- written instructions for safely joining and leaving the traffic stream when entering and leaving the location of the business, residence, or institution
- contact information for the Prime Contractor

2. Planning and Operations

The plan to use a pilot car should be assessed before the project commences, taking into account the nature of the work zone. For example, if there are complex access issues, a strategy should be developed for keeping track of vehicle entries and departures from the queue.

A pilot car shall be operated in a manner that ensures the highest level of safety for road users and workers.

The travel speed should not permit gaps to develop between the vehicles being led, and should not contribute to tar splatter or the creation of dust in the work zone.

Traffic Control Persons and pilot car operators should remain in radio communication throughout the work zone, with Traffic Control Persons regulating traffic:

- at each end of the work zone
- at every intersection that may require it between Traffic Control Persons
- at every other location where needed to ensure safety
- when an assessment dictates it, at every business access location that routinely has customers stopping between the primary Traffic Control Persons at either end of the pilot car zone

3. Pilot Car Warning Lights

4-way flashers and 360-degree rotating yellow warning lights shall be used on pilot cars. Warning lights should be directly wired to the vehicle's electrical system. Arrow sticks are an acceptable alternative to the 360-degree rotating yellow warning lights.

The lights should be used only when the pilot car is operating.

4. Pilot Car Signs

The Pilot Car C-048-1-DS sign is usually double-sided and should have the words PILOT CAR on one side for approaching vehicles and PILOT CAR – DO NOT PASS on the other side for following vehicles. The sign should be positioned on the pilot car so that it is visible to drivers of vehicles approaching from both directions, and shall be kept in a vertical position to ensure good viewing from both directions.

Pilot cars may use two separate signs with the same messaging indicated above. Another option is the Pilot Car C-048-2 sign, which displays PILOT CAR on both sides and fits within manufactured overhead racks.

Shoulder-mounted FOLLOW PILOT CAR signs should be positioned so that approaching drivers are notified of the pilot car operation. Additional FOLLOW PILOT CAR signs may be required so that signs are positioned beyond the end of the expected or known vehicle queues resulting from pilot car operations.



C-049-x
900 x 900 mm



C-048-1-DS
1200 x 900 mm



C-048-2
1830 x 305 mm



Figure 4.34: Pilot Car Signs

5. Pilot Car Radio Communications

Pilot cars should have an electronic device that allows all pilot car operators and all Traffic Control Persons to communicate effectively with each other over the length of the pilot car operation.

Communications should be on the same channel or frequency, and should be switched on at all times during pilot car operations.

6. Pilot Car Traffic Control

Pilot car operators are not authorized to direct traffic. Their role is to guide traffic through a work zone once that traffic has been directed to follow them by an authorized Traffic Control Person. Pilot car operators may also position their vehicles to control vehicle access to the work zone.

If a pilot car operator is to carry out Traffic Control Person operations, that operator shall first be trained as a Traffic Control Person in a manner that is acceptable to WorkSafeBC, and shall perform Traffic Control Person duties in a manner that complies with WorkSafeBC and Road Authority requirements.

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PART B – TRAFFIC CONTROL

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PART B – TRAFFIC CONTROL

Section 5: Traffic Control Persons (TCPs)

5.1 Roles and Responsibilities

For information on management responsibilities related to traffic control, see *Section 1.2: Road Authority and Prime Contractor Responsibilities.*

5.1.1 Traffic Control Supervisor

The Prime Contractor shall designate a Traffic Control Supervisor who is qualified to assume the responsibilities of this function. It cannot be the Site Supervisor, Superintendent, or Foreman unless the designation is authorized by the Road Authority.

The Traffic Control Supervisor may be an employee of the Prime Contractor or a sub-contractor to the Prime Contractor. It may be the Traffic Control Manager for the project or an onsite Traffic Control Person if circumstances allow. If the traffic management responsibilities for the project require full-time or frequent attention, a different person should be assigned to this role.

In general, the Traffic Control Supervisor is responsible for the following:

- oversee traffic control operations, ensuring traffic control is executed in accordance with the Traffic Control Plan, and updated as necessary.
- ensure compliance with Part 18 of WorkSafeBC's Occupational Health and Safety Regulation regarding supervision of Traffic Control Persons in the work zone
- provide direction to Traffic Control Persons
- required traffic control devices are in place
- signs are checked, maintained, and moved as required
- daily traffic control setups are documented, and changes are identified in the Traffic Control Plan or log book
- traffic concerns are reported to the Traffic Control Manager or Site Supervisor
- each member of the traffic control crew wears the required personal protective clothing and equipment (see *Section 5.4: Work Zone Apparel and Equipment*)

The Traffic Control Supervisor shall also ensure that all TCPs are:

- carrying evidence of current TCP certification
- equipped with all necessary equipment, including, radios, spare batteries, chargers, and red signalling wands
- performing traffic control duties competently and safely
- positioned in safe locations that are clear of potential environmental hazards, such as a slide or avalanche
- provided with rest breaks

If two or more TCPs work as a team, the employer of the traffic control personnel and the Traffic Control Supervisor should ensure that the responsibility for coordinating changes in traffic flow is assigned appropriately. The Traffic Control Supervisor shall have TCP certification in order to assume the duties of a TCP and direct traffic.

5.1.2 Traffic Control Persons (TCPs)

Depending on the project category and complexity, and in collaboration with the Prime Contractor, TCPs may be required to prepare, review, amend, and document Traffic Control Plans as part of their daily activities. In order to do so, they shall carry valid TCP certification on the work site at all times, and have a good working knowledge of this Manual.

TCPs shall communicate instructions and directions to drivers effectively by using standard traffic control motions and signals that are precise and deliberate to be clearly understood by road users.

TCPs quickly become familiar with their work zone, and should try to assess the layout through the eyes of a road user who is arriving at the zone in the worst foreseeable conditions. This will help them to anticipate traffic control issues and identify required changes to the Traffic Control Plan.

TCP training and performance should emphasize:

- the importance of the job
- alertness and attentiveness
- the need for a courteous but firm manner
- proper TCP positioning in relation to the work activity area in order to achieve effective traffic control and ensure the safety of the public, workers, equipment, and all TCPs
- sufficient discipline to prevent others from loitering near the TCP location
- sufficient discipline to remain in position until relieved by other personnel or until the conflict being controlled no longer exists
- the requirement to remove or cover Traffic Control Person Ahead C-001-1 signs whenever TCPs are not actively regulating traffic
- the requirement to comply with Part 18 of WorkSafeBC's Occupational Health and Safety Regulation regarding personal protective equipment (see *Section 5.4: Work Zone Apparel and Equipment*) and traffic control signals (see *Section 5.6: TCP Positioning and Signals*)

Among other requirements specified in Part 18 of WorkSafeBC's Occupational Health and Safety Regulation, employers and supervisors should ensure that:

- Traffic control arrangements and procedures for the work are made known to all personnel involved in the work.
- Required traffic control devices and procedures are in place before the work starts and are removed when they are no longer required.
- Any person assigned to be a TCP is adequately trained in a manner acceptable to WorkSafeBC, and performs effectively in accordance with the traffic control arrangements and procedures for the work.

5.2 Use of TCPs in Work Zones

TCPs are used only when all other traffic control methods are considered inadequate to warn, direct, and regulate road users within a work zone.

TCPs are used to direct traffic within a work zone, thereby preventing conflicts between the movements of pedestrians, vehicles, workers, and work zone equipment.

TCPs shall not control traffic within speed limits greater than 70 km/h.

Common applications for TCPs include:

- control of alternating one-way traffic through sections of a two-way road that is temporarily reduced to one lane (single lane alternating traffic or SLAT)
- stopping public traffic to permit equipment to cross or enter onto a road, structure, or other work zone feature
- providing information to drivers or pedestrians regarding road closures, wait times, route options, etc.

One or both TCPs may be omitted in the following situations:

1. The self-regulated section does not extend through an intersection.
2. Where an open, one-lane section is sufficiently short (e.g., a spot obstruction), sight distance is adequate, and traffic volumes are light.
    - If one TCP is omitted, the Traffic Control Person Ahead C-001-1 signs remain to warn of the one TCP ahead.
    - If both TCPs are omitted, the Traffic Control Person Ahead C-001-1 signs are removed and a Yield To Oncoming Traffic R-056 sign is posted in the closed lane or the lane affected by the works.
3. A temporary traffic signal may be used instead of TCPs to control traffic on sections of one-lane, two-way roadways (see *Section 4.8: Temporary and Portable Traffic Signals*).
4. TCPs are not generally required as a traffic control measure for reducing speed or for reducing the number of lanes on multilane roads.